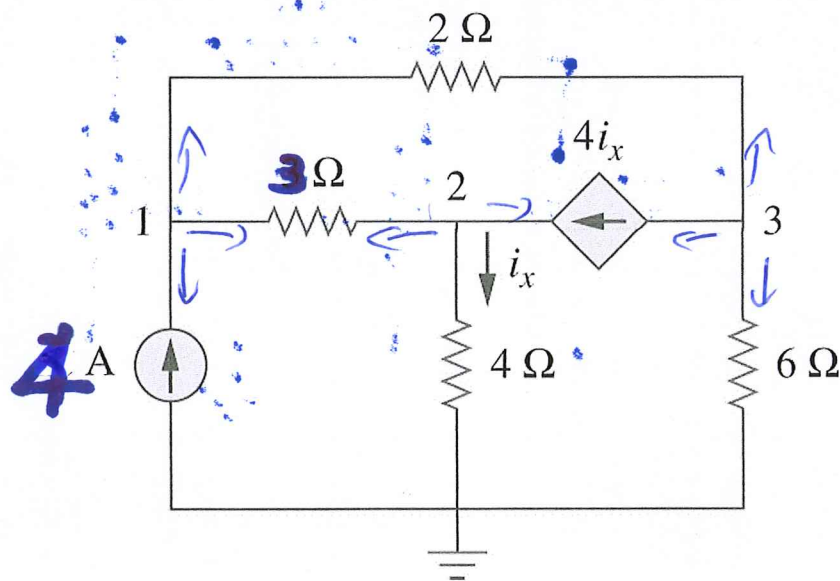


2) Determine the node voltages.



$$N_1: \frac{V_1 - V_2}{3} + \frac{V_1 - V_3}{2} - 4 = 0$$

$$2V_1 - 2V_2 + 3V_1 - 3V_3 = 24$$

$$5V_1 - 2V_2 - 3V_3 = 24 \quad \text{--- ①}$$

$$N_2: \frac{V_2 - V_1}{3} + \frac{V_2}{4} - \frac{4V_2}{4} = 0$$

$$\times 12: 4V_2 - 4V_1 + 3V_2 - 12V_2 = 0$$

$$-4V_1 - 5V_2 + 0V_3 = 0 \quad \text{--- ②}$$

$$N_3: \frac{V_3 - V_1}{2} + \frac{V_3}{6} + \frac{4V_2}{4} = 0$$

$$\times 12: 6V_3 - 6V_1 + 2V_3 + 12V_2 = 0$$

$$-6V_1 + 12V_2 + 8V_3 = 0 \quad \text{--- ③}$$

Method 1:

Q2

From (2):  $V_2 = -\frac{4V_1}{5} = -0,8V_1 \dots (4)$

Sub (4) in (1):  $5V_1 - 2(-0,8V_1) - 3V_3 = 24$

$$5V_1 + 1,6V_1 - 3V_3 = 24$$

$$6,6V_1 - 3V_3 = 24 \dots (5)$$

Sub (4) in (3):  $-6V_1 + 12(-0,8V_1) + 8V_3 = 0$

$$-6V_1 - 9,6V_1 + 8V_3 = 0$$

$$-15,6V_1 + 8V_3 = 0 \dots (6)$$

(5)  $\times 8$ :  $52,8V_1 - 24V_3 = 192 \dots (7)$

(6)  $\times 3$ :  $-46,8V_1 + 24V_3 = 0 \dots (8)$

(7) + (8):  $6V_1 = 192$

$$V_1 = 32 \text{ V}$$

$$V_3 = \frac{15,6V_1}{8} = 62,4 \text{ V}$$

$$V_2 = -0,8V_1 = -25,6 \text{ V}$$

Method 2:

G2

$$\begin{bmatrix} 5 & -2 & -3 \\ -4 & -5 & 0 \\ -6 & 12 & 8 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 24 \\ 0 \\ 0 \end{bmatrix}$$

$$\Delta = \begin{bmatrix} 5 & -2 & -3 \\ -4 & -5 & 0 \\ -6 & 12 & 8 \end{bmatrix} = \begin{aligned} & (5)(-5)(8) + (-4)(12)(-3) + (-6)(-2)(0) \\ & - (-3)(-5)(-6) - (0)(12)(5) - (8)(-2)(-4) \end{aligned}$$

$$= -200 + 144 + 0 + 90 - 0 - 64$$

$$= -30$$

$$\Delta_1 = \begin{bmatrix} 24 & -2 & -3 \\ 0 & -5 & 0 \\ 0 & 12 & 8 \end{bmatrix} = \begin{aligned} & (24)(-5)(8) + 0 + 0 \\ & = 0 - 0 - 0 \end{aligned}$$

$$= -960$$

$$\Delta_2 = \begin{bmatrix} 5 & 24 & -3 \\ -4 & 0 & 0 \\ -6 & 0 & 8 \end{bmatrix} = \begin{aligned} & 0 + 0 + 0 - 0 - 0 \\ & - (8)(24)(-4) \end{aligned}$$

$$= 768$$

$$\Delta_3 = \begin{bmatrix} 5 & -2 & 24 \\ -4 & -5 & 0 \\ -6 & 12 & 0 \end{bmatrix} = \begin{aligned} & 0 + (-4)(12)(24) + 0 \\ & = - (24)(-5)(-6) - 0 - 0 \end{aligned}$$

$$= -1152 - 720$$

$$= -1872$$

$$v_1 = \frac{\Delta_1}{\Delta} = 32 \text{ V}$$

$$v_2 = \frac{\Delta_2}{\Delta} = -25,6 \text{ V}; \quad v_3 = \frac{\Delta_3}{\Delta} = 62,4 \text{ V}$$